Error Messages & Validation

In short

Vague error messages, or fields that only turn red, can stop people finishing a form. Say exactly what's wrong and how to fix it, and don't rely on colour. Link each message to its field, and move focus to the first error.

What is it for?

Form errors are a major barrier for all users, but they can completely block users with disabilities. Inaccessible errors include:

- Just highlighting a field in red is useless for screen reader users and people with colour blindness
- Vague messages like 'Invalid input' or 'Error' tell the user nothing about how to fix it
- When a form fails to submit, screen reader users are often left at the top of the page, not knowing what or where the error is

This is a core design pattern, not a single tag. Support depends on using standard HTML and Accessible Rich Internet Applications (ARIA) attributes (`aria-describedby`, `aria-invalid`, `aria-live`) correctly. This is a key part of the 'Perceivable', 'Understandable', and 'Robust' principles of the Web Content Accessibility Guidelines (WCAG), and works in all modern browsers.

How to support it

Errors need to be clear, visible, and announced.

- Be specific. Don't say 'Invalid', instead say something like 'Please enter a valid email address (for example, name@example.com).'
- Don't use colour alone, always add a text message, an accessible icon (like an X), or both, next to the field.
- Connect the error. Use `aria-describedby` to programmatically link the input field to its error message. This tells screen readers

to announce the error when the user focuses on that field
- Manage Focus. When a form submission fails, ideally, move focus to the first field with an error. If that's not possible, move the focus to a summary of errors at the top of the form
- Use aria-invalid="true". Set this on the input field so assistive technology knows it has an error

Useful links

- World Wide Web Consortium (W3C) WAI - Validating Input
- Level Access - Designer Tips: Creating Accessible Forms

Abbreviations

The full form of each abbreviation used in this document.

ARIA: Accessible Rich Internet Applications

HTML: HyperText Markup Language

W3C: World Wide Web Consortium

WAI: Web Accessibility Initiative

WCAG: Web Content Accessibility Guidelines